



FULL LENGTH MOCK TEST #03 (BEGINNERS)

Time: 90mins

Marks: 300

[1]: . What is the distance between the two foci of the hyperbola $25x^2 - 75y^2 = 225$?

Option 1:

$2\sqrt{3}$ units

Option 2:

$4\sqrt{3}$ units

Option 3:

$\sqrt{6}$ units

Option 4:

$2\sqrt{6}$ units

SolutionHeading: Full Solution

Text:

Calculation:

Given,

Hyperbola equation: $25x^2 - 75y^2 = 225$

Divide both sides by 225 to obtain standard form:

$$\frac{25x^2}{225} - \frac{75y^2}{225} = 1 \implies \frac{x^2}{9} - \frac{y^2}{3} = 1$$

Thus, $a^2 = 9$ and $b^2 = 3$.

Compute c from $c^2 = a^2 + b^2$:

$$c^2 = 9 + 3 = 12 \implies c = \sqrt{12} = 2\sqrt{3}.$$

The foci are at $(\pm c, 0)$, so the distance between them is $2c$:

$$2c = 2 \times 2\sqrt{3} = 4\sqrt{3}.$$

$\therefore$ The distance between the two foci is $4\sqrt{3}$ units.

Hence, the correct answer is Option 2.

[2]: The sum of the first 20 terms of the series $5 + 11 + 19 + 29 + 41 + \dots$ is

Option 1:

3450

Option 2:

3250

Option 3:

3420

Option 4:

3520

SolutionHeading: Full Solution

Text:

Calculation:

$$S_{20} = 5 + 11 + 19 + 29 + \dots$$

$$\text{Let } T_r = ar^2 + br + c$$

$$\Rightarrow T_1 = a + b + c = 5$$

$$\Rightarrow T_2 = 4a + 2b + c = 11$$

$$\Rightarrow T_3 = 9a + 3b + c = 19$$

$$a = 1, \quad b = 3, \quad c = 1$$

$$\text{Hence } S_{20} = \sum_{r=1}^{20} r^2 + 3 \sum_{r=1}^{20} r + \sum_{r=1}^{20} 1 = 3520$$

Hence, the correct answer is Option 4.

[3]: Let A and B be two sets containing 4 and 2 elements respectively. Then the number of subsets of the set $A \times B$, each having at least three elements is

Option 1:

219

Option 2:

235

Option 3:

228

Option 4:

256

SolutionHeading: Full Solution

Text:

Given, $n(A) = 4, n(B) = 2$

$\therefore n(A \times B) = 8$

Thus the number of subsets of $A \times B$ having at least 3 elements

$$= 2^8 - {}^8C_0 - {}^8C_1 - {}^8C_2 = 256 - 1 - 8 - 28 = 219$$

Positive Marks: 4.00

Negative Marks: 1.00

[4]: The Boolean expression for the following would be:

$$F(A, B, C) = \Sigma(0, 2, 6)$$

$$d(A, B, C) = \Sigma(1, 3, 5)$$

Option 1:

 $A'C' + BC'$

Option 2:

 $B + AC'$

Option 3:

 $A + BC'$

Option 4:

 $A' + BC'$

SolutionHeading: Full Solution

Text:

The correct answer is **$A' + BC'$**

 **Key Points** A Karnaugh Map (K-map) is a visual method used for simplifying Boolean expressions. It provides a straightforward way to minimize expressions with two, three, four, or more variables. The map is a grid where each cell represents a min-term of the Boolean function. By organizing these min-terms in a specific pattern, it becomes easier to spot opportunities for simplification.

Don't-care terms (denoted as (x)) are used in Boolean functions to indicate that for certain combinations of variables, the output value can be either 0 or 1 without affecting the overall functionality of the circuit or logic. These terms provide flexibility for further simplification and optimization of the Boolean expression.

	$\overline{B}\overline{C}$	$\overline{B}C$	BC	$B\overline{C}$
$\overline{A}$	1	X	X	1
A		X		1

$$\Rightarrow (A'B'C' + A'B'C + A'BC + A'BC') + (A'BC' + ABC')$$

$$\Rightarrow A(B'C' + B'C + BC + BC') + BC'(A' + A)$$

$$\Rightarrow A' + BC'$$

So, the correct answer is $A' + BC'$

Positive Marks: 4.00

Negative Marks: 1.00

[5]: A process execute the following code:

For ($I = 1; I \leq n; I++$) fork ();

How many child processes are created?

Option 1:

$n - 1$

Option 2:

$2^n - 1$

Option 3:

2^{n-1}

Option 4:

$2^{n-1} - 1$

SolutionHeading: Full Solution

Text:

Concept:

Fork () is a system call that creates a new process. After new child process is created, both processes will execute the next instruction. If fork() returns a negative value, creation of child process was unsuccessful. Return zero to newly created process.

Explanation:

If there are n `fork()` calls then number of child processes created are $2^n - 1$.

Example:

For ($i = 1; i \leq 3; i++$) `fork ()`;

Here $n = 3$,

Number of child processes created = $2^3 - 1 = 7$

Positive Marks: 4.00

Negative Marks: 1.00

[6]: D flip flop can be made from a J-K flip flop by making

Option 1:

$J = K$

Option 2:

$J = K = 1$

Option 3:

$J = 0, K = 1$

Option 4:

$J = \bar{K}$

SolutionHeading: Full Solution

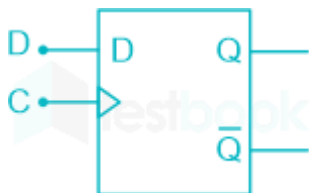
Text:

D flip flop:

D flip flop has only one input terminal. **The output of the D flip flop will be the same as the input.** Hence, it is used in delay circuits.

The circuit is as shown below.

Logic symbol:



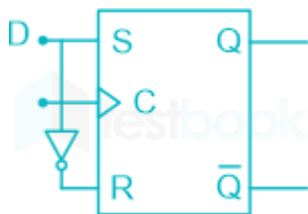
Truth table:

D	Q_n (Present state)	Q_{n+1} (Next state)
0	0	0
0	1	0
1	0	1
1	1	1

Characteristic equation: $Q_{n+1} = D$

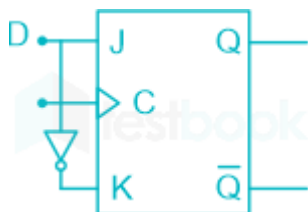
The **D flip flop** may be obtained from an **S-R flip flop** by just putting one **inverter between the S and R** as shown in the figure below.

$$S = \bar{R}$$



The **D flip flop** may be obtained from a **J-K flip flop** by just putting one **inverter between the J and K** as shown in the figure below.

$$K = \bar{J}$$

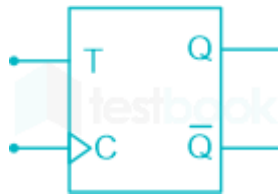


T flip flop:

T flip flop has only one input terminal. **The output of the T flip flop will be toggled when the input is**

high on every new clock pulse. The output will be the same as the previous state when the input is low. The circuit is as shown below.

Logic symbol:

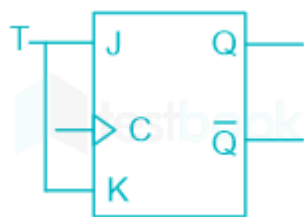


Truth table:

T	Q_n (Present state)	Q_{n+1} (Next state)
0	0	0
0	1	1
1	0	1
1	1	0

Characteristic equation: $Q_{n+1} = T\bar{Q}_n + \bar{T}Q_n$

The **T flip flop** may be obtained from a **J-K flip flop** by making both the inputs are the same i.e. **J = K**.



Positive Marks: 4.00

Negative Marks: 1.00

[7]: Page fault occurs when:

Option 1:

the page is corrupted by application software

Option 2:

the page is in main memory

Option 3:

the page is not in main memory

Option 4:

one tries to divide a number by 0

SolutionHeading: Full Solution

Text:

The correct answer is **the page is not in main memory**.

Key Points

- A **page fault** occurs when a program tries to access a block of memory (a page) that is not currently stored in the computer's main memory (RAM).
- When this happens, the operating system must retrieve the page from a secondary storage device, such as a hard drive, which can cause a delay in program execution.
- This is a normal part of memory management in systems that use virtual memory, and it allows the system to use more memory than is physically available by swapping pages in and out of the main memory.

Positive Marks: 4.00

Negative Marks: 1.00

[8]: The worst-case time complexity of accessing an element in a singly linked list is _____.

Option 1:

Option 2:

Option 3:

Option 4:

$\Theta(1)$ $O(n)$ $O(1)$ $O(n^2)$

SolutionHeading: Full Solution

Text:

The correct answer is **option 2**.

Concept:

Singly-linked list:

A singly linked list is a particular kind of linked list that can only be traversed in one direction, from head to final node (tail). A node is a name given to each entry in a linked list. Data and a pointer to the following node are both included in a single node, which aids in keeping the list's structure.

The head node, which points to the list's root node and facilitates access to all other elements, is the first node. We can tell when the list terminates because the last node, usually known as the tail, links to NULL.

The worst case Time Complexity for performing these operations is as follows:

- Add item to the front of the list: $O(1)$.
- Accessing an element in a singly linked list is $O(n)$.
- Add item to the end of the list: $O(n)$.
- Add item to anywhere in the list: $O(n)$.

Explanation:

Accessing an element in a singly linked list is $O(n)$, because a linked list has n nodes we can not go directly to the n -th element. So we have to move the entire list of n nodes so it takes $O(n)$ time.

Hence the correct answer is $O(n)$.

[9]: If two lists are given such as:

L1: [2, 4, 6, 8, 10]

L2: [10, 4, 6, 8, 10]

If the lists are sorted using Insertion sort then which of the lists List1 or List 2 will make the minimum number of comparisons?

Option 1:

L1

Option 2:

L2

Option 3:

Same for L1 and L2.

Option 4:

No comparisons required

SolutionHeading: Full Solution

Text:

Insertion sort works by the algorithm like to compare every element with the element in its left side , if they are smaller , skip , else swap the element.

Applying insertion sort in list 1 :

L1- {2,4,6,8,10}

If the list is already sorted , then insertion sort do (n-1) number of comparison .

So, in total we need 4 comparison for this List .

L2 - {10,4,6,8,10}

Pass 1 - compare 10 and 4 , swap the position -

{4,10,6,8,10} - 1 comparison

Pass 2 - compare 10 and 6 , swap them , again compare 6 and 4(no swapping required -

{4,6,10,8,10} - 2 comparison

Pass 3 - Compare 8 and 10 , swap them , again compare 8 with 6 and 4(no swap as they are already sorted) - {4,6,8,10,10}-3 comparison

Pass 4 - compare 10 with 10 , as they are same

element , so, no swapping required ,and here , list got sorted - {4,6,8,10,10} - 1 comparison
So, in total we need 7 comparison to sort list 2 .

Question asks about minimum number of comparison , so, Option 1 will be the answer.

Positive Marks: 4.00

Negative Marks: 1.00

[10]: If you randomly select a letter from the word "ELEPHANT," what is the probability that it is a vowel?

Option 1:

3/4

Option 2:

1/3

Option 3:

3/8

Option 4:

1/2

SolutionHeading: Full Solution
Text:

Given:

Word = "ELEPHANT"

Total letters = 8

Vowels = {E, E, A} = 3 vowels

Formula used:

Probability = (Number of favorable outcomes) / (Total outcomes)

Calculations:

Probability = (Number of vowels) / (Total letters)

⇒ Probability = 3 / 8

∴ The correct answer is option (3).

Positive Marks: 4.00

Negative Marks: 1.00

[11]: In a G.P. consisting of positive terms, each term equals the sum of the next two terms. Then, the common ratio of this progression equals:

Option 1:

$$\frac{1}{2}(\sqrt{5} - 1)$$

Option 2:

$$\frac{1}{2}(1 - \sqrt{5})$$

Option 3:

$$\frac{1}{2}\sqrt{5}$$

Option 4:

$$\sqrt{5}$$

SolutionHeading: Full Solution

Text:

Calculation:

Let a be the first term and r be the common ratio of the G.P.

The relationship given is:

$$T_n = T_{n+1} + T_{n+2}$$

Using the general formula $T_n = a \times r^{n-1}$:

$$a \times r^{n-1} = a \times r^n + a \times r^{n+1}$$

Divide the entire equation by the non-zero term $a \times r^{n-1}$ (since all terms are positive, $a \neq 0$ and $r \neq 0$):

$$\Rightarrow 1 = \frac{ar^n}{ar^{n-1}} + \frac{ar^{n+1}}{ar^{n-1}}$$

$$\Rightarrow 1 = r + r^2$$

Rearrange the equation into a standard quadratic form $ar^2 + br + c = 0$:

$$\Rightarrow r^2 + r - 1 = 0$$

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow r = \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times (-1)}}{2 \times 1}$$

$$\Rightarrow r = \frac{-1 \pm \sqrt{1+4}}{2}$$

$$\Rightarrow r = \frac{-1 \pm \sqrt{5}}{2}$$

Since the G.P. consists of positive terms, the common ratio r must be positive ($r > 0$).

The two possible values for r are:

$$r_1 = \frac{-1+\sqrt{5}}{2} \text{ (Positive, since } \sqrt{5} \approx 2.236 > 1)$$

$$r_2 = \frac{-1-\sqrt{5}}{2} \text{ (Negative, since the numerator is negative)}$$

The required common ratio is the positive root r_1 :

$$\Rightarrow r = \frac{\sqrt{5}-1}{2}$$

$$\Rightarrow r = \frac{1}{2}(\sqrt{5} - 1)$$

Positive Marks: 4.00

Negative Marks: 1.00

[12]: A bag contains 5 white and 4 red balls. A ball is drawn at random from the bag. What is the probability that the ball drawn is **not** red?

Option 1:

$$\frac{4}{9}$$

Option 2:

$$\frac{5}{9}$$

Option 3:

$$\frac{1}{3}$$

Option 4:

$$\frac{2}{3}$$

SolutionHeading: Full Solution

Text:

Given:

Number of white balls = 5 Number of red balls = 4

Concept used:

The probability of an event occurring is given by the number of favorable outcomes divided by the total number of possible outcomes.

Calculation:

Total number of balls = Number of white balls +
Number of red balls

$$= 5 + 4 = 9$$

Number of favorable outcomes (drawing a ball that is not red) = Number of white balls

= 5

Probability = Number of favorable outcomes / Total number of possible outcomes

= 5 / 9

Therefore, the probability that the ball drawn is not red is 5/9.

The correct option is: 2) 5/9

Positive Marks: 4.00

Negative Marks: 1.00

[13]: Consider the following statements about

the matrix $M = \begin{bmatrix} 71 & 23 & 48 \\ 57 & 28 & 29 \\ 65 & 17 & 48 \end{bmatrix}$

Statement-I: The inverse of M does not exist.

Statement-II: M is non-singular.

Which one of the following is correct in respect of the above statements?

Option 1:

Both Statement-I and Statement-II are correct and Statement-II explains Statement-I

Option 2:

Both Statement-I and Statement-II are correct but Statement-II does not explain Statement-I

Option 3:

Statement-I is correct but Statement-II is not correct

Option 4:

Statement-I is not correct but Statement-II is correct

Statement-I is not correct but Statement-II is correct

SolutionHeading: Full Solution

Text:

Calculation:

$$\text{Given the matrix } M = \begin{pmatrix} 71 & 23 & 48 \\ 57 & 28 & 29 \\ 65 & 17 & 48 \end{pmatrix}$$

Determinant of Matrix M

The inverse of a matrix exists if the determinant of the matrix is non-zero. So, we calculate the determinant of matrix M using the formula for the determinant of a 3x3 matrix:

$$\det(M) = 71 \times \begin{vmatrix} 28 & 29 \\ 17 & 48 \end{vmatrix} - 23 \times \begin{vmatrix} 57 & 29 \\ 65 & 48 \end{vmatrix} + 48 \times \begin{vmatrix} 57 & 28 \\ 65 & 17 \end{vmatrix}$$

After calculating the individual 2x2 determinants:

$$\det(M) = 71 \times 851 - 23 \times 851 + 48 \times (-851)$$

$$\det(M) = 60421 - 19573 - 40848 = 0$$

Since the determinant of the matrix M is 0, this means the matrix M is **singular**, and as a result, the inverse of M does not exist.

Statement-I: "The inverse of M does not exist" is correct.

Statement-II: "M is non-singular" is incorrect because M is singular ($\det(M) = 0$).

Therefore, the correct answer is:

Statement-I is correct but Statement-II is not correct.

Hence, the correct answer is Option 3.

[14]: A set contains $2n + 1$ elements. The number of subsets of this set containing more than n elements is equal to

Option 1:

$$2^{n-1}$$

Option 2:

$$2^n$$

Option 3:

$$2^{n+1}$$

Option 4:

$$2^{2n}$$

SolutionHeading: Full Solution

Text:

Explanation:

Let the original set contains $(2n + 1)$ elements, then subsets of this set containing more than n elements, i.e., subsets containing $(n + 1)$ elements, $(n + 2)$ elements,..... $(2n + 1)$ elements.

Required number of subsets

$$= {}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + \dots + {}^{2n+1}C_{2n} + {}^{2n+1}C_{2n+1}$$

$$= {}^{2n+1}C_n + {}^{2n+1}C_{n-1} + \dots + {}^{2n+1}C_1 + {}^{2n+1}C_0$$

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_{n-1} + {}^{2n+1}C_n$$

$$= \frac{1}{2}[(1 + 1)^{2n+1}] = \frac{1}{2}[2^{2n+1}] = 2^{2n}$$

∴ The number of subsets of this set containing more than n elements is equal to 2^{2n} .

[15]: With reference to computer memory, which of the following Register is also known as Memory Buffer Register?

Option 1:

Main Address Register

Option 2:

Memory Data Registers

Option 3:

Memory Address Register

Option 4:

Primary Data Register

SolutionHeading: Full Solution

Text:

The correct answer is Memory Data Register

Key Points

- **Register memory is the smallest and fastest memory in a computer and is not a part of the main memory.**
 - It is located in the CPU in the form of registers, which are the smallest data-holding elements.
 - A register temporarily holds frequently used data, instructions, and memory addresses that are to be used by the CPU.
 - **A memory Data Register (MDR) is a register used for holding the information that is in the process of being transferred from the memory to the central processor.**
 - **It is also called Memory Buffer Register.**
- Hence option 2 is correct.**
- Memory Data Register contains a copy of the value in the memory location specified by the Memory Address Register (MAR).

Additional Information

- **Memory Address Register or MAR is a temporary storage in CPU or a parallel loading register that contains the next memory address to be read or written.**
- **Contents of the MAR are directly connected to the address bus and are used to handle the**

address transferred to the memory unit.

- When reading from memory, data addressed by MAR is stored in the Memory Data Register (MDR) and is used by the CPU.
- **When writing to memory, the CPU writes data from MDR to the memory location whose address is stored in MAR.**

Positive Marks: 4.00

Negative Marks: 1.00

[16]: What is the primary difference between ROM and RAM?

Option 1:

ROM is faster than RAM

Option 2:

ROM is volatile, while RAM is non - volatile

Option 3:

ROM is used for data storage, while RAM is used for program execution

Option 4:

ROM stores permanent data that cannot be changed, while RAM stores temporary data that can be read and written

SolutionHeading: Full Solution

Text:

The correct answer is **ROM stores permanent data that cannot be changed, while RAM stores temporary data that can be read and written**

Key Points

- **ROM (Read-Only Memory)** is a type of non-volatile memory that **stores permanent data**,

such as firmware and essential system instructions.

- The data in **ROM is written during the manufacturing process**, and it is typically not modified or changed during normal operation.
- In contrast, **RAM (Random Access Memory)** is a type of **volatile memory** that is used for temporary storage of data that can be both read and written.
- RAM is used for actively running programs and storing temporary data that is needed by the computer while it is powered on.
- Unlike ROM, the data in **RAM is lost when the power is turned off.**

Positive Marks: 4.00

Negative Marks: 1.00

[17]: Match the following:

1	Longest common subsequence	i	Divide and conquer
2	Huffman code	ii	Linear programming
3	Quick sort	iii	Greedy algorithm
4	Simplex algorithm	iv	Dynamic programming

Option 1:

1 - iv, 2 - i, 3 - ii, 4 - iii

Option 3:

1 - iii, 2 - iv, 3 - ii, 4 - i

Option 2:

1 - iii, 2 - ii, 3 - iv, 4 - i

Option 4:

1 - iv, 2 - iii, 3 - i, 4 - ii

SolutionHeading: Full Solution

Text:

Longest Common Subsequence - Dynamic Programming

- Dynamic Programming: Dynamic programming is an algorithmic paradigm that solves problems by

breaking them down into subproblems and storing the results of these subproblems to avoid redundant computation. The longest common subsequence problem can be solved using dynamic programming by breaking it down into subproblems and building the solution for the larger problem based on the solutions to the smaller subproblems.

Huffman Code - Greedy Algorithm

- Greedy algorithms make locally optimal choices at each step of the algorithm in the hope of finding a global optimum. Huffman code is a compression algorithm that constructs a variable-length prefix code based on the frequency of occurrence of each symbol in the input data. This can be done using a greedy algorithm that constructs the prefix code by repeatedly merging the two least frequent symbols.

Quick Sort - Divide and Conquer

- Divide and conquer is an algorithmic paradigm that solves problems by breaking them down into smaller subproblems, solving the subproblems recursively, and combining the solutions to solve the original problem. Quick sort is a sorting algorithm that sorts an array by partitioning it into two subarrays, sorting the subarrays recursively, and then combining the sorted subarrays.

Simplex Algorithm - Linear Programming

- Linear programming is a method to optimize a linear objective function subject to linear constraints. The simplex algorithm is an algorithm

that solves linear programming problems by iteratively moving from one feasible solution to a neighboring feasible solution that improves the objective function until an optimal solution is reached. The simplex algorithm works by constructing a simplex, which is a polytope of feasible solutions, and moving from one vertex of the simplex to an adjacent vertex that improves the objective function.

Positive Marks: 4.00

Negative Marks: 1.00

[18]: Which of the following is the type of a linked list?

Option 1:

single linked list

Option 2:

doubly linked list

Option 3:

circular linked list

Option 4:

All of the above

SolutionHeading: Full Solution

Text:

Types of the linked list:

1. Single linked list:

This list consists of a number of nodes in which each node has a next pointer to the following element. The link of the last node in the list is NULL, which indicates the end of the list.

2. Doubly linked list:

It is a two-way linked list. A user can navigate in both directions. In a doubly linked list, the user can delete a node even if he doesn't have the previous node's address.

3. Circular linked list:

In singly linked lists and doubly linked lists, the end of

lists is indicated with a NULL value. But circular linked lists do not have ends. In circular linked lists, each node has a successor and last node points to the first node instead of NULL.

Positive Marks: 4.00

Negative Marks: 1.00

[19]: Match the following.

List I	List II
I. Stack Algorithm	P. Deadlock
II. Elevator Algorithm	Q. Disk Scheduling
III. Priority Scheduling Algorithm	R. Page Replacement
IV. Havender's Algorithm	S. CPU Scheduling

Option 1:

I – R, II – Q, III – S, IV – P

Option 2:

I – Q, II – R, III – S, IV – P

Option 3:

I – R, II – Q, III – P, IV – S

Option 4:

I – Q, II – R, III – P, IV – S

SolutionHeading: Full Solution

Text:

Stack algorithm:

A page replacement algorithm is said to satisfy the inclusion property, or it is called as stack algorithm if the set of pages in a k frame memory is always a subset of the pages in a k+1 frame memory.

Elevator algorithm:

It is a disk scheduling algorithm, also known as scan algorithm in which head, starts from one end of disk and moves towards other end by servicing each request in between and reach other end.

Priority scheduling algorithm:

It is a CPU scheduling algorithm in which each process is assigned a priority and processes according to their priority.

Havender's Algorithm:

It is related with deadlock. In this, all the resources required by process will be requested at once.

Positive Marks: 4.00

Negative Marks: 1.00

[20]: What is number of page faults by least recently used page replacement for a memory with 4 frames for the page reference string 2, 0, 1, 2, 4, 0, 5, 1, 4, 6, 4, 2, 1, 3, 0 ?

Option 1:

8

Option 2:

7

Option 3:

9

Option 4:

10

SolutionHeading: Full Solution

Text:

Concept:

Least Recently Used (LRU) is a page replacement technique that replaces the least recently used page first from the frame in case of a page miss.

Hit/Miss Table:

String	2	0	1	2	4	0	5	1	4	6	4	2	1	3	0
Frames	2	2	2	0	0	1	2	4	0	5	5	1	6	4	2
		0	0	1	1	2	4	0	5	1	1	6	4	2	1
			1	2	2	4	0	5	1	4	6	4	2	1	3
					4	0	5	1	4	6	4	2	1	3	0
Hit/Miss	M	M	M	H	M	H	M	M	H	M	H	M	H	M	M

M → Miss

H → Hit

Since the number of miss is 10. Therefore, number of page fault is 10.

Positive Marks: 4.00

Negative Marks: 1.00

[21]: Consider the following set of processes and the length of CPU burst time given in milliseconds:

Process	CPU Burst time (ms)
P ₁	5
P ₂	7
P ₃	6
P ₄	4

Assume that processes being scheduled with Round-Robin Scheduling Algorithm with Time

Quantum 4ms. Then the waiting time for P₄ is _____ ms.

Option 1:

0

Option 2:

4

Option 3:

12

Option 4:

6

SolutionHeading: Full Solution

Text:

Scheduling Algorithm: Round-Robin

Time Quantum: 4 milliseconds

Gantt chart:

P ₁	P ₂	P ₃	P ₄	P ₁	P ₂	P ₃
----------------	----------------	----------------	----------------	----------------	----------------	----------------

0 4 8 12 16 17 20 22

Process Table:

Process	Arrival Time (AT)	Burst Time (BT)	Completion Time (CT)	Turnaround Time (TAT)	Waiting Time (WT)
P ₁	0	5	17	17	12
P ₂	0	7	20	20	13
P ₃	0	6	22	22	16
P ₄	0	4	16	16	12

Therefore, the waiting time for P₄ is 12 milliseconds.

Formula used:

$TAT = CT - AT$

$WT = TAT - BT$

Important Points:

Since arrival time is not given of processes is not given consider all the process arrive at time = 0 and by convention, the process with lowest id associated with it executes first, that is, P_1 followed by P_2 and so on.

Positive Marks: 4.00

Negative Marks: 1.00

[22]: Which of the following are not shared by the threads of the same process?

- (a) Stack
- (b) Registers
- (c) Address Space
- (d) Message Queue

Option 1:

(a) and (d)

Option 2:

(b) and (c)

Option 3:

(a) and (b)

Option 4:

(a), (b) and (c)

SolutionHeading: Full Solution

Text:

- A thread is a lightweight process that can be managed independently by a scheduler.
- A message queue is a linked list of messages stored within the kernel and identified by a message queue identifier.
- Threads share the code and data segments and the heap, but they don't share the stack and registers. Message queue comes under data and therefore it is shared by thread.
- Threads of the same process don't share stack and register, hence option 3 is correct

Positive Marks: 4.00

Negative Marks: 1.00

[23]: Match List-I with List-II

List – I	List – II
(a) Disk	(i) Thread
(b) CPU	(ii) Signal
(c) Memory	(iii) File System
(d) Interrupt	(iv) Virtual address

Choose the correct option from those given below:

Option 1:

(a)-(i); (b)-(ii); (c)-(iii); (d)-(iv)

Option 2:

(a)-(iii); (b)-(i); (c)-(iv); (d)-(ii)

Option 3:

(a)-(ii); (b)-(i); (c)-(iv); (d)-(iii)

Option 4:

(a)-(ii); (b)-(iv); (c)-(iii);(d)-(i)

SolutionHeading: Full Solution

Text:

Concept:

- File system is the part of the operating system which is responsible for file management. It provides a mechanism to store the data and access to the file contents including data and programs. The File system takes care of the following issues: File Structure, Recovering Free

space, Disk space assignment to the files and Tracking data location.

- Threads refer to the highest level of code executed by a processor, so with many threads, your CPU can handle several tasks at the same time. All CPUs have active threads, and every process performed on your computer has at least a single thread.
- A virtual address space (VAS) or address space is the set of ranges of virtual addresses that an operating system makes available to a process.
- an interrupt is an input signal to the processor indicating an event that needs immediate attention. An interrupt signal alerts the processor and serves as a request for the processor to interrupt the currently executing code, so that the event can be processed in a timely manner.

Conclusion:

Disk → File system, CPU → Thread, memory → Virtual address space and Interrupt → Signal

Therefore option 2 is correct

Positive Marks: 4.00

Negative Marks: 1.00

[24]: Find the median of the given set of numbers 4, 6, 3, 8, 5, 2, 7, 9

Option 1:

6.5

Option 2:

5.5

Option 3:

5

Option 4:

6

SolutionHeading: Full Solution

Text:

Concept:

Median: The median is the middle number in a sorted- ascending or descending list of numbers.

Case 1: If the number of observations (n) is even

$$\text{Median} = \frac{\text{value of } \left(\frac{n}{2}\right)^{\text{th}} \text{ observation} + \text{value of } \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ observation}}{2}$$

Case 2: If the number of observations (n) is odd

$$\text{Median} = \text{value of } \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

Calculation:

Given values 4, 6, 3, 8, 5, 2, 7, 9

Arrange the observations in ascending order:

2, 3, 4, 5, 6, 7, 8, 9

Here, $n = 8 = \text{even}$

As we know, If n is even then,

$$\begin{aligned}\text{Median} &= \frac{\text{value of } \left(\frac{n}{2}\right)^{\text{th}} \text{ observation} + \text{value of } \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ observation}}{2} \\ &= \frac{4^{\text{th}} \text{ observation} + 5^{\text{th}} \text{ observation}}{2} \\ &= \frac{6+5}{2} = 5.5\end{aligned}$$

Hence Median = 5.5

Positive Marks: 4.00

Negative Marks: 1.00

[25]: If $|x^2 - 3x + 2| > x^2 - 3x + 2$, then which one of the following is correct?

Option 1:

$$x \leq 1 \text{ or } x \geq 2$$

Option 2:

$$1 \leq x \leq 2$$

Option 3:

$$1 < x < 2$$

Option 4:

x is any real value except 3 and 4

SolutionHeading: Full Solution

Text:

Concept:

$$\text{If } f(x) = |x|, \text{ then } f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Calculation:

$$\text{Given: } |x^2 - 3x + 2| > x^2 - 3x + 2$$

Case-1: If $x \geq 0$

$$\Rightarrow x^2 - 3x + 2 > x^2 - 3x + 2$$

$\therefore$ No real value of x satisfies the above equation.

Case-2:- If $x < 0$

$$\Rightarrow -(x^2 - 3x + 2) > x^2 - 3x + 2$$

$$\Rightarrow x^2 - 3x + 2 < 0$$

$$\Rightarrow (x - 1)(x - 2) < 0$$

$$\Rightarrow 1 < x < 2.$$

Positive Marks: 4.00

Negative Marks: 1.00

[26]: The circle $x^2 + y^2 + 4x - 7y + 12 = 0$ cut an intercept on y- axis equal

Option 1:

1

Option 2:

3

Option 3:

4

Option 4:

7

SolutionHeading: Full Solution

Text:

Concept:

The Intercept made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on:

I. X-axis is given by: $2\sqrt{g^2 - c}$

II. Y-axis is given by: $2\sqrt{f^2 - c}$

Note: Intercepts are always positive.

Calculation:

Given: Equation of circle

$$x^2 + y^2 + 4x - 7y + 12 = 0$$

Now, by comparing the given equation of circle with the standard equation of circle $x^2 + y^2 + 2gx + 2fy + c = 0$, we get:

$$g = 2, f = -7/2 \text{ and } c = 12.$$

As we know that, the intercept made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on Y – axis is given by:

$$2\sqrt{f^2 - c}.$$

$$\Rightarrow 2\sqrt{f^2 - c} = 2\sqrt{\left(-\frac{7}{2}\right)^2 - 12} = 1$$

Positive Marks: 4.00

Negative Marks: 1.00

[27]: If $f(x)$ and $g(x)$ are continuous functions satisfying $f(x) = f(a - x)$ and $g(x) + g(a - x) = 2$, then what is $\int_0^a f(x) g(x) dx$ equal to?

Option 1:

$$\int_0^a g(x) dx$$

Option 2:

$$\int_0^a f(x) dx$$

Option 3:

$$2 \int_0^a f(x) dx$$

Option 4:

$$0$$

SolutionHeading: Full Solution

Text:

Concept:

Integral Properties:

- $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

Calculation:

Given: $f(x) = f(a - x)$ & $g(x) + g(a - x) = 2$

To find: $\int_0^a f(x) g(x) dx$

$$I = \int_0^a f(x) g(x) dx \quad \dots (1)$$

Using integral property, $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

$$\Rightarrow I = \int_0^a f(a - x) g(a - x) dx$$

Using given, $f(x) = f(a - x)$

$$\Rightarrow I = \int_0^a f(x) g(a - x) dx \quad \dots (2)$$

Adding equation (1) & (2)

$$2I = \int_0^a f(x) g(x) dx + \int_0^a f(x) g(a-x) dx$$

$$\Rightarrow 2I = \int_0^a f(x) \{g(x) + g(a-x)\} dx$$

Using given, $g(x) + g(a-x) = 2$

$$\Rightarrow 2I = \int_0^a f(x) \times 2 dx$$

$$\Rightarrow I = \int_0^a f(x) dx$$

Positive Marks: 4.00

Negative Marks: 1.00

[28]: Consider the following statements:

1. Derivative of $f(x)$ may not exist at some point.
2. Derivative of $f(x)$ may exist finitely at some point.
3. Derivative of $f(x)$ may be infinite (geometrically) at some point.

Which of the above statements are correct?

Option 1:

1 and 2 only

Option 2:

2 and 3 only

Option 3:

1 and 3 only

Option 4:

1, 2 and 3

SolutionHeading: Full Solution

Text:

Calculation:

Let's analyze each statement

1. Let, $f(x) = |x|$,

$$f(x) = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

$$f'(x) = \begin{cases} -1, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

Left hand derivative $\neq$ Right hand derivative

At $x = 0$, $f'(x)$ doesn't exist, so statement 1 is true.

2. Let, $f(x) = \tan^{-1}(x)$

Now, $f'(x) = 1/(1 + x^2) = \text{finite value}$ (At some points)

So Derivative of $f(x)$ may exist finitely at some point.

Hence statement 2 is also true

3. We know that derivative of function is undefined at some points it means derivative of $f(x)$ may be infinite (geometrically) at some point.

So, statement 3 is also true

Hence, option (4) is correct.

Positive Marks: 4.00

Negative Marks: 1.00

[29]: Let A and B be subsets of X and $C = (A \cap B') \cup (A' \cap B)$, where A' and B' are complements of A and B respectively in X . what is C equal to?

Option 1:

$$(A \cup B') - (A \cap B')$$

Option 3:

$$(A \cup B) - (A \cap B)$$

Option 2:

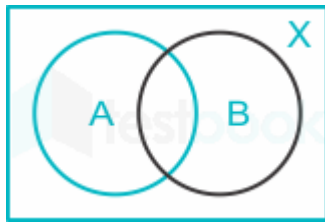
$$(A' \cup B) - (A \cap B)$$

Option 4:

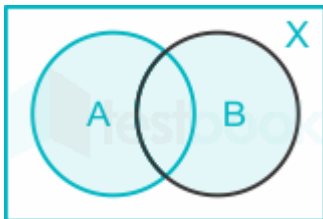
$$(A' \cup B') - (A' \cap B')$$

Concept:

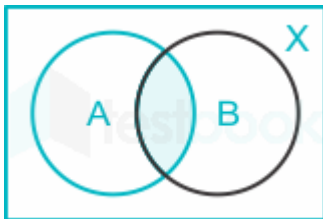
Let A and B be the subsets of X, then then **Venn Diagram** to represent that is:



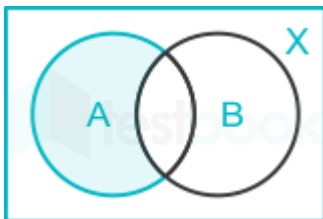
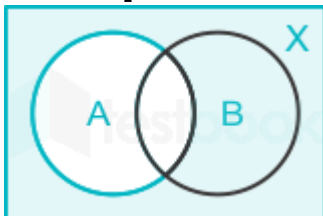
- **Union of sets ($A \cup B$):**



- **Intersection of subsets ($A \cap B$):**

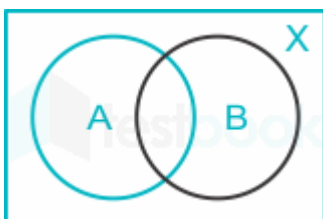


- **Complement of subset (A'):**

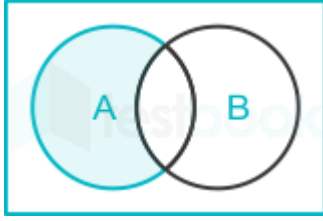


Calculation:

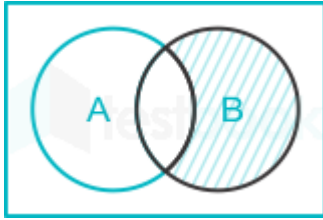
Let figure given below denotes A and B as subsets of X.



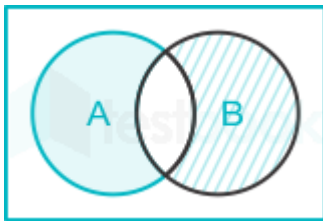
Now, $(A \cap B')$ is A intersection not B i.e., $(A \cap B') = A - (A \cap B)$ as shown below.



Similarly, $(A' \cap B)$ is B intersection not A i.e., $(A' \cap B) = B - (A \cap B)$ as shown below.



Now, $C = (A \cap B') \cup (A' \cap B)$ means $(A \cup B) - (A \cap B)$ as shown below.



- **Difference of subsets $(A - B)$:**
-

Positive Marks: 4.00

Negative Marks: 1.00

[30]: Consider the following statements:

1. $f(x) = [x]$, where $[.]$ is the greatest integer function, is discontinuous at $x = n$, where $n \in \mathbb{Z}$.
2. $f(x) = \cot x$ is discontinuous at $x = n\pi$, where $n \in \mathbb{Z}$.

Which of the above statements is/are correct?

Option 1:

Option 2:

Option 3:

Option 4:

1 only

2 only

Both 1 and 2

Neither 1 nor 2

SolutionHeading: Full Solution

Text:

Concept:

- The greatest integer function is discontinuous at all the integers.
- For function to be continuous it should be first defined

Calculation:

Statement (1) is correct as the greatest integer function is discontinuous at all the integers.

$$2. f(x) = \cot x$$

$$\Rightarrow f(x) = \cos x / \sin x$$

$$\text{At } x = n\pi, \sin x = 0$$

$\therefore \cot x$ is not defined at $x = n\pi$

Hence, $\cot x$ is discontinuous at $x = n\pi$

So this statement (2) is correct.

Hence, option (3) is correct.

Positive Marks: 4.00

Negative Marks: 1.00

[31]: What is $\int \frac{dx}{x(1+\ln x)^n}$ equal to ($n \neq 1$) ?

Option 1:

$$\frac{1}{(n-1)(1+\ln x)^{n-1}} + c$$

Option 2:

$$\frac{1-n}{(1+\ln x)^{1-n}} + c$$

Option 3:

$$\frac{n+1}{(1+\ln x)^{n+1}} + c$$

Option 4:

$$-\frac{1}{(n-1)(1+\ln x)^{n-1}} + c$$

SolutionHeading: Full Solution

Text:

Concept:

$$\int x^a dx = \frac{x^{a+1}}{a+1} + C$$

Calculation:

$$\int \frac{dx}{x(1+\ln x)^n}$$

Let us consider $1 + \ln(x) = t$.

Now by differentiating both the sides with respect to x , we get

$$\Rightarrow \frac{d(1+\ln x)}{dx} = \frac{dt}{dx} \Rightarrow \frac{dx}{x} = dt \quad \text{-----(1)}$$

Now by using the equation (1), we get

$$\Rightarrow \int \frac{dx}{x(1+\ln x)^n} = \int \frac{dt}{t^n} = \int t^{-n} dt$$

$$\text{As we know that, } \int x^a dx = \frac{x^{a+1}}{a+1} + C$$

$$\Rightarrow \int \frac{dx}{x(1+\ln x)^n} = \int t^{-n} dt = \frac{t^{1-n}}{1-n} + C$$

Now by substituting $1 + \ln(x) = t$ in the above equation we get,

$$\Rightarrow \int \frac{dx}{x(1+\ln x)^n} = \frac{t^{1-n}}{1-n} + C = -\frac{1}{(n-1) \times (1+\ln x)^{n-1}} + C$$

Positive Marks: 4.00

Negative Marks: 1.00

[32]: The eccentricity of the hyperbola $16x^2 - 9y^2 = 1$ is

Option 1:

Option 2:

Option 3:

Option 4:

$$\frac{3}{5}$$

$$\frac{5}{3}$$

$$\frac{4}{5}$$

$$\frac{5}{4}$$

SolutionHeading: Full Solution

Text:

Concept:

Hyperbola: The locus of a point which moves such that its distance from a fixed point is greater than its distance from a fixed straight line. (Eccentricity = $e > 1$)

Calculation:

Given:

$$16x^2 - 9y^2 = 1$$

$$\Rightarrow \frac{x^2}{\frac{1}{16}} - \frac{y^2}{\frac{1}{9}} = 1$$

$$\text{Compare with } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\therefore a^2 = 1/16 \text{ and } b^2 = 1/9$$

$$\text{Eccentricity} = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{\left(\frac{1}{9}\right)}{\left(\frac{1}{16}\right)}} = \sqrt{1 + \frac{16}{9}} = \sqrt{\frac{25}{9}} = \frac{5}{3}$$

Positive Marks: 4.00

Negative Marks: 1.00

[33]: The number of ways in which a cricket team of 11 players can be chosen out of a batch of 15 players so that the captain of the team is always included, is

Option 1:

165

Option 2:

364

Option 3:

1001

Option 4:

1365

SolutionHeading: Full Solution

Text:

Concept:

Suppose a word of n letters has n_1 of one kind of letter, n_2 of a second kind, n_3 of a third kind, and so on with $n = n_1 + n_2 + n_3 + \dots + n_k$ then the number of distinguishable permutations of the n letters is =

$$\frac{n!}{n_1! \times n_2! \times n_3! \times \dots \times n_k!}$$

The number of ways of selecting r objects from n different objects subject to certain conditions like:

1. k particular objects are always included $= {}^{n-k}C_{r-k}$
2. k particular objects are never included $= {}^nC_r$
3. **Calculation:**

We know that The number of ways of selecting r objects from n different objects subject to a certain condition is:

4. k particular objects are always included $= {}^{n-k}C_{r-k}$
 5. $\therefore$ Total number of selection of 11 players out of 15 players in which the captain is included =
- $${}^{14}C_{10} = 1001$$

Positive Marks: 4.00

Negative Marks: 1.00

[34]: If a line is perpendicular to the line $5x - y = 0$ and forms a triangle of area 5 square units with co-ordinate axes, then its equation is

Option 1:

$$x + 5y \pm 5\sqrt{2} = 0$$

Option 3:

$$5x + y \pm 5\sqrt{2} = 0$$

Option 2:

$$x - 5y \pm 5\sqrt{2} = 0$$

Option 4:

$$5x - y \pm 5\sqrt{2} = 0$$

SolutionHeading: Full Solution

Text:

Concept:

- Equation of a line **perpendicular to a given line** $ax + by + c = 0$ is $bx - ay + \lambda = 0$, where λ is a constant.
- The **intercept form of the line** $\Leftrightarrow \frac{x}{a} + \frac{y}{b} = 1$, Where a is the x - intercept and b is the y - intercept.

Calculation:

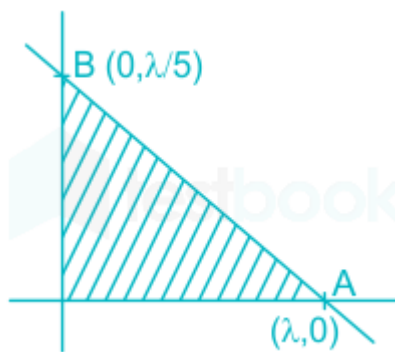
Given: Area of triangle is 5 square units

Given equation of line is $5x - y = 0$

Equation of a line perpendicular to a given line is

$$\Rightarrow x + 5y = \lambda \quad \dots (1)$$

$$\Rightarrow \frac{x}{\lambda} + \frac{5y}{\lambda} = 1 \Rightarrow \frac{x}{\lambda} + \frac{y}{\left(\frac{\lambda}{5}\right)} = 1$$



Area of triangle = 5

$$\Rightarrow \frac{1}{2} \times \lambda \times \frac{\lambda}{5} = 5$$

$$\Rightarrow \lambda^2 = 50$$

$$\therefore \lambda = \pm 5\sqrt{2}$$

Put the value of λ in equation 1st

$$\text{So, } x + 5y = \pm 5\sqrt{2}$$

$$\Rightarrow x + 5y \pm 5\sqrt{2} = 1$$

[35]:

If $G(x) = \sqrt{25 - x^2}$ then what is $\lim_{x \rightarrow 1} \frac{G(x) - G(1)}{x - 1}$ equal to?

Option 1:

$$-\frac{1}{2\sqrt{6}}$$

Option 2:

$$\frac{1}{5}$$

Option 3:

$$-\frac{1}{\sqrt{6}}$$

Option 4:

$$\frac{1}{\sqrt{6}}$$

SolutionHeading: Full Solution

Text:

Concept:

L-Hospital Rule: Let $f(x)$ and $g(x)$ be two functions

Suppose that we have one of the following cases,

$$\text{I. } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$$

$$\text{II. } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\infty}{\infty}$$

Then we can apply L-Hospital Rule $\Leftrightarrow \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

Calculation:

$$\lim_{x \rightarrow 1} \frac{G(x) - G(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{\sqrt{(25-x^2)} - \sqrt{(25-1)}}{x - 1}$$

Here, we have 0/0 form, so applying L-Hospital Rule,

$$\lim_{x \rightarrow 1} \frac{\sqrt{(25-x^2)} - \sqrt{(25-1)}}{x - 1} = \lim_{x \rightarrow 1} \frac{\frac{1(-2x)}{2\sqrt{(25-x^2)}}}{1}$$

$$= \lim_{x \rightarrow 1} \frac{-x}{\sqrt{(25-x^2)}}$$

$$= -\frac{1}{\sqrt{24}}$$

$$= -\frac{1}{2\sqrt{6}}$$

Hence, option (a) is correct.

Positive Marks: 4.00

Negative Marks: 1.00

What is the value of $\left(\frac{-1+i\sqrt{3}}{2}\right)^{3n} + \left(\frac{-1-i\sqrt{3}}{2}\right)^{3n}$?

Where $i = \sqrt{-1}$?

Option 1:

3

Option 2:

2

Option 3:

1

Option 4:

0

SolutionHeading: Full Solution

Text:

Concept:

Cube Roots of unity are 1, ω and ω^2

Here, $\omega = \frac{-1+i\sqrt{3}}{2}$ and $\omega^2 = \frac{-1-i\sqrt{3}}{2}$

Property of cube roots of unity

1. $\omega^3 = 1$

2. $1 + \omega + \omega^2 = 0$

3. $\omega = 1 / \omega^2$ and $\omega^2 = 1 / \omega$

4. $\omega^{3n} = 1$

Calculation:

We know that $\omega = \frac{-1+i\sqrt{3}}{2}$ and $\omega^2 = \frac{-1-i\sqrt{3}}{2}$

$$\left(\frac{-1+i\sqrt{3}}{2}\right)^{3n} + \left(\frac{-1-i\sqrt{3}}{2}\right)^{3n} = \omega^{3n} + (\omega^2)^{3n} = (\omega^3)^n + (\omega^3)^{2n} = 1 + 1 = 2$$

Positive Marks: 4.00

Negative Marks: 1.00

[37]: The sum of the focal distances of a point on an ellipse is constant and equal to the

Option 1:

length of minor axis

Option 2:

length of major axis

Option 3:

length of latus rectum

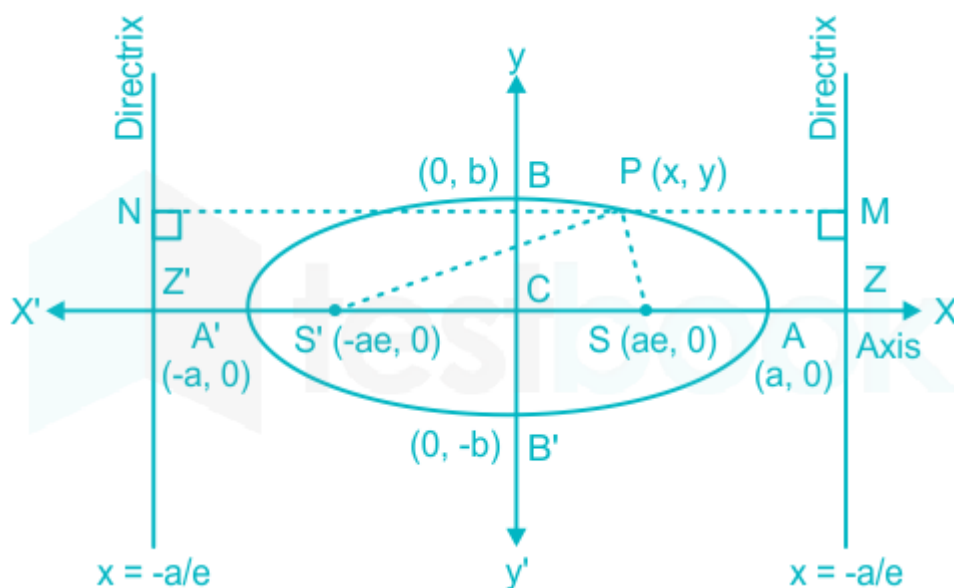
Option 4:

sum of the lengths of semi-major and semi-minor axes

SolutionHeading: Full Solution

Text:

Explanation:



From the figure and the property of ellipse

$$PM = (a/e) - x$$

$$\text{And, } SP = e \times PM$$

$$\Rightarrow SP = e \times [(a/e) - x]$$

$$\Rightarrow SP = a - ex \quad \text{----(i)}$$

Similarly from the figure and

$$PN = (a/e) + x$$

$$\text{And, } S'P = e \times PN$$

$$\Rightarrow S'P = e \times [(a/e) + x]$$

$$\Rightarrow S'P = a + ex \quad \text{----(ii)}$$

From equation (i) and (ii) we get

$$SP + S'P = a - ex + a + ex$$

$$\Rightarrow SP + S'P = 2a \text{ (Length of major axis)}$$

★ **Important Points** In the case of vertical ellipse it must be equal to the length of the minor axis.

Positive Marks: 4.00

Negative Marks: 1.00

[38]: If the lines $3y + 4x = 1$, $y = x + 5$ and $5y + bx = 3$ are concurrent, then what is the value of b ?

Option 1:

1

Option 2:

3

Option 3:

6

Option 4:

1/2

SolutionHeading: Full Solution

Text:

Concept:

If $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are 3 lines then these lines are said to be concurrent if:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Calculation:

Given: the lines $3y + 4x = 1$, $y = x + 5$ and $5y + bx = 3$ are concurrent.

Now, by comparing the three lines with the standard equation of line $ax + by + c = 0$ we get:

$$\Rightarrow a_1 = 4, b_1 = 3, c_1 = -1, a_2 = 1, b_2 = -1, c_2 = 5, a_3 = b, b_3 = 5 \text{ and } c_3 = -3.$$

As we know that, if $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are 3 lines then these lines are said to be concurrent if:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

$$\begin{vmatrix} 4 & 3 & -1 \\ 1 & -1 & 5 \\ b & 5 & -3 \end{vmatrix} = 14b - 84 = 0$$

$$\Rightarrow b = 6.$$

Positive Marks: 4.00

Negative Marks: 1.00

[39]:

$$\text{Consider the function } f(x) = \begin{cases} ax - 2 & \text{for } -2 < x < -1 \\ -1 & \text{for } -1 \leq x \leq 1 \\ a + 2(x - 1)^2 & \text{for } 1 < x < 2 \end{cases}$$

What is the value of a for which f(x) is continuous at x = -1 and x = 1?

Option 1:

-1

Option 2:

1

Option 3:

0

Option 4:

2

SolutionHeading: Full Solution

Text:

Concept:

Continuity of a function: A function f(x) is said to be continuous at point x = a if the following criterion holds: $\lim_{x \rightarrow a^-} f(x) = f(a) = \lim_{x \rightarrow a^+} f(x)$

Calculation:

First observe that f(1) = f(-1) = -1.

Now consider the left-hand limit for x = -1:

$$\lim_{x \rightarrow -1^-} (ax - 2) = -a - 2$$

Now equating with right-hand limit and the functional value we get:

$$\begin{aligned} -a - 2 &= -1 \\ 1 - 2 &= a \\ a &= -1 \end{aligned}$$

Similarly consider the right-hand limit for $x = 1$:

$$\lim_{x \rightarrow 1^+} (a + 2(x - 1)^2) = a$$

Now equating with left-hand limit and the value of the function we get:

$$a = -1$$

therefore, the function is continuous at $x = -1$ and $x = 1$ for $a = -1$.

Positive Marks: 4.00

Negative Marks: 1.00

[40]: Find the next number in the following series:

14, 31, 59, 121, 239, ?

Option 1:

490

Option 2:

484

Option 3:

481

Option 4:

478

SolutionHeading: Full Solution

Text:

The logic followed here is:



So, '481' is the next number in the given series.

Hence, the correct answer is "Option 3".

Positive Marks: 4.00

Negative Marks: 1.00

[41]: Minimum number of 2 : 1 multiplexer required to design a 16 : 1 multiplexer is _____

Option 1:

8

Option 2:

4

Option 3:

15

Option 4:

10

SolutionHeading: Full Solution**Text:**

To implement $2n \times 1$ MUX using 2×1 MUX, the total number of 2×1 MUX required is $(2n - 1)$.

$\therefore$ The number of 2×1 multiplexer required to implement 16×1 MUX will be:

$$n = 16 - 1 = 15$$

Or we can follow the below steps to calculate the same:

$$1^{\text{st}} \text{ stage} = \frac{16}{2} = 8$$

$$2^{\text{nd}} \text{ stage} = \frac{8}{2} = 4$$

$$3^{\text{rd}} \text{ stage} = \frac{4}{2} = 2$$

$$4^{\text{th}} \text{ stage} = \frac{2}{2} = 1$$

The sum will give the total number of MUX required to implement 16×1 multiplexer using 2×1 , i.e.

$$n = 8 + 4 + 2 + 1 = 15$$

**Important Point**

Given MUX	To be implemented	Required
2 : 1	16 : 1	$8 + 4 + 2 + 1 = 15$
4 : 1	16 : 1	$4 + 1 = 5$
4 : 1	64 : 1	$16 + 4 + 1 = 21$
8 : 1	64 : 1	$8 + 1 = 9$
8 : 1	256 : 1	$32 + 4 + 1 = 37$

Positive Marks: 4.00

Negative Marks: 1.00

[42]: Excess 3 code of decimal 3 is :

Option 1:

100

Option 2:

1010

Option 3:

0001

Option 4:

0110

SolutionHeading: Full Solution

Text:

The correct answer is **0110**.

Key Points

- **Excess-3** or **XS3** is a **non-weighted BCD** (Binary Coded Decimal) code used for decimal digits. It is also termed as biased representation.
- It is called excess-3 because it starts at binary equivalent of decimal number 3.
- To convert a decimal number to excess-3 code, you **add 3** to the decimal number and then convert that sum into binary equivalent.
- For example to convert decimal 3 to excess-3 code,
 - First, add 3 to the decimal number: **$3+3 = 6$**
 - Then, convert that sum into binary equivalent: 6(in decimal) = 0110(in binary)

So, the Excess-3 code of decimal 3 is 0110. This corresponds to option (4).

Positive Marks: 4.00

Negative Marks: 1.00

[43]: In Kernel I/O Subsystem, A _____ is a buffer that holds jobs for a device until it is ready to take them.

Option 1:

Buffering

Option 2:

Caching

Option 3:

Spooling

Option 4:

Scheduling

SolutionHeading: Full Solution

Text:

The correct answer is **option 3**.

Concept:

Many I/O services are provided by the kernel. Caching, scheduling, spooling, device reservation, and error handling are all tasks provided by the kernel that rely on the hardware and device driver infrastructure.

- A **spooling** is a type of buffer that holds the jobs for a system until the system is ready to accept the jobs.
- Spooling considers discs to be a huge buffer capable of holding as many jobs as the device requires until the output devices are ready to accept them.
- Spooling allows a user to view and, if desired, delete specific data streams.

Hence the correct answer is Spooling.

Additional Information Caching:

Caching involves storing a replica of data in a location that is easier to reach than the original.

Buffering:

The buffer is a section of main memory used to temporarily store or keep data sent between two devices or between a device and an application.

Positive Marks: 4.00

Negative Marks: 1.00

[44]: The main function of the dispatcher (the portion of the process scheduler) is:

Option 1:

swapping a process to the disk

Option 2:

assigning ready process to the CPU

assigning ready process to the CPU

Option 3:

suspending some of the processes when the CPU load is high

Option 4:

bring processes from the disk to the main memory

SolutionHeading: Full Solution

Text:

The correct answer is assigning ready process to the CPU.

Key Points

- The dispatcher is a crucial component of the process scheduler in an operating system.
- Its primary role is to assign a ready process to the CPU, ensuring efficient process management and CPU utilization.
- When a process is in the ready state, the dispatcher switches it to the running state by loading its context into the CPU.
- This context switch involves saving the state of the current process and loading the state of the next process.



Additional Information

- **Swapping a process to the disk:** This is a function of memory management, not the dispatcher. It involves moving a process from main memory to disk (swap space) to free up memory.

- **Suspending some of the processes when the CPU load is high:** This is part of process management and load balancing but not specifically the dispatcher's role. It involves moving processes to a suspended state to manage CPU load.
- **Bring processes from the disk to the main memory:** This is also a memory management function, often handled by the swapper or pager, not the dispatcher.

Positive Marks: 4.00

Negative Marks: 1.00

[45]: Consider the following segment table:

Segment	Base	Length
0	227	598
1	2200	15
2	92	102
3	1325	550
4	1943	100

Calculate the physical address of the logical address: <1, 10>

NOTE: <a, b> means <segment number, segment offset>

Option 1:

2210

Option 2:

2225

Option 3:

2205

Option 4:

Invalid

SolutionHeading: Full Solution

Text:

Segmentation: This technique solves the problem of paging where the user's view of memory allocation is

not followed. Instead of dividing the memory into equal-sized pages, virtual address space is divided into variable-length segments. This suffers from external fragmentation.

Segment Table: This table maintains an entry for each segment in logical address space. Each entry of the segment table contains the base and the limit.

- **Limit:** It specifies the length of the segment.
- **Base:** It specifies the starting address of the segment that resides in the memory.

Calculation:

1. In the above table, the logical address is $\langle 1, 10 \rangle$.
2. In the segment table, for this logical address, the base is 2200 and the length is 15.
3. The offset in logical address should be strictly less than the limit (length) specified in the segment table
4. Hence, $10 < 15$.
5. Now, Physical Address = Base + Offset
 $\Rightarrow 2200 + 10$
 $\Rightarrow 2210$

Positive Marks: 4.00

Negative Marks: 1.00

[46]: Match List - I with List - II.

LIST - I		LIST - II	
A.	Complement	I.	Adder
B.	Ex-OR	II.	Instruction decoding
C.	Accumulator	III.	Negative number representation
D.	Control Unit	IV.	Arithmetic and Logic operations

Choose the correct answer from the options given below :

Option 1:

A - I, B - II, C - III, D - IV

Option 2:

A - III, B - II, C - I, D - IV

Option 3:

A - III, B - I, C - IV, D - II

Option 4:

A - IV, B - II, C - III, D - I

SolutionHeading: Full Solution

Text:

The correct answer is **A - III, B - I, C - IV, D - II**

Key Points

. A. Complement

- Complement is often used in the representation of negative numbers, particularly in binary systems.
- Correct match: III. Negative number representation

. B. Ex-OR

- Ex-OR (Exclusive OR) gate is commonly used in adder circuits for binary addition.
- Correct match: I. Adder

. C. Accumulator

- An accumulator is a register in a CPU that stores intermediate arithmetic and logic results.
- Correct match: IV. Arithmetic and Logic operations

• **D. Control Unit**

- The control unit is responsible for decoding instructions in a CPU and coordinating the execution of these instructions.
- Correct match: II. Instruction decoding

Therefore, the correct answer is: A - III, B - I, C - IV, D - II

Positive Marks: 4.00

Negative Marks: 1.00

[47]: A two-digit number is four times the sum and three times the product of its digits. Find the number.

Option 1:

22

Option 2:

24

Option 3:

26

Option 4:

28

SolutionHeading: Full Solution

Text:

Given:

A two-digit number is four times the sum and three times the product of its digits.

Formula Used:

Let the two-digit number be $10a + b$, where a and b are its digits.

Equation 1: $10a + b = 4(a + b)$

Equation 2: $10a + b = 3ab$

Calculation:

From Equation 1:

$$10a + b = 4(a + b)$$

$$\Rightarrow 10a + b = 4a + 4b$$

$$\Rightarrow 10a - 4a = 4b - b$$

$$\Rightarrow 6a = 3b$$

$$\Rightarrow 2a = b$$

From Equation 2:

$$10a + b = 3ab$$

Substitute $b = 2a$ into Equation 2:

$$10a + 2a = 3a \times 2a$$

$$\Rightarrow 12a = 6a^2$$

$$\Rightarrow 2 = a$$

Substitute $a = 2$ into $b = 2a$:

$$b = 2 \times 2$$

$$\Rightarrow b = 4$$

So, the two-digit number is:

$$10a + b = 10 \times 2 + 4$$

$$\Rightarrow 24$$

The number is 24.

Positive Marks: 4.00

Negative Marks: 1.00

[48]: In a stack, we can access the element-

Option 1:

All of the options

Option 2:

From any position

Option 3:

Which is entered at last

Option 4:

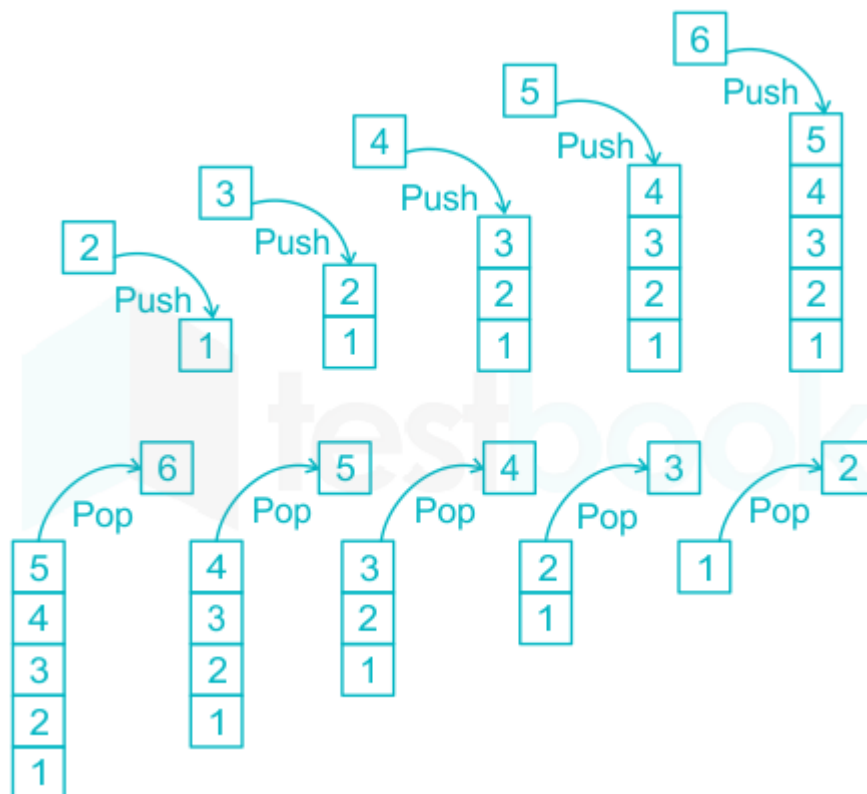
Which is entered at beginning

The correct answer is **Which is entered at last**

Key Points

- Stack is **LIFO** type data structure. **LIFO** stands for Last-in-first-out.
- Here, the element which is placed (inserted or added) last, is accessed first.
- In stack terminology, insertion operation is called PUSH operation and removal operation is called POP operation.
- We can see push(insertion) and POP(removal) both are occurring at top of stack.

In the given example we have inserted “6” in the last but removed “6” first. This is LIFO operation.



Positive Marks: 4.00

Negative Marks: 1.00

[49]: What is the worst-case and average-case time complexity of the Binary search?

Option 1:

$$O(n^2)$$

Option 2:

$$O(1)$$

Option 3:

$$O(n \log n)$$

Option 4:

$$O(\log n)$$

SolutionHeading: Full Solution

Text:

Binary search algorithm:

Binary search algorithm is used to find an element in an already sorted array.

STEPS 1:

It finds the middle element of the array and compare the element with the element to be searched, if it matches then return true.

STEPS 2:

If not, then divide the array into two halves in which if element to be search is less than middle element then search takes place in left part otherwise search in right half.

STEP 3:

Repeat this process, until you get the element.

Explanation:

For worst case 52

Worst Case: Search 50 in below give array

11	12	15	24	35	50	51	63
----	----	----	----	----	----	----	----

$$\text{midde index} = \frac{0+9}{2} = 4 \therefore a[4] = 35$$

$$50 > 32$$

$$\text{midde index} = \frac{5+9}{2} = 7 \therefore a[7] = 63$$

$$50 < 63$$

$$\text{midde index} = \frac{5+6}{2} = 8 \therefore a[5] = 50$$

matched

$$T(n) = O(\log n)$$

Also, for average case:

$$T(n) = O(\log n)$$

[50]: Which of the followings are sequential circuit?

- A. Flip flop
- B. Multiplexer
- C. Counter
- D. Decoder
- E. Adder

Choose the correct answer from the options given below :

Option 1:

B, D Only

Option 2:

A, C Only

Option 3:

C, E Only

Option 4:

D, A Only

SolutionHeading: Full Solution

Text:

The correct answer is **2) A, C Only**.

Key Points

- **Sequential circuits** are types of circuits where the output depends not just on the current inputs, but also on the history of inputs. This means they have memory elements that store previous states.
- **Flip flop** is a basic memory element used in sequential circuits, which can store one bit of data.
- **Counter** is a sequential circuit that goes through a prescribed sequence of states upon the application of input pulses.

Additional Information

- **Multiplexer (MUX)** is a combinational circuit that selects one of many input signals and forwards the selected input into a single line. It does not store any state information, hence it is not a sequential circuit.

- **Decoder** is also a combinational circuit that converts binary information from n input lines to a maximum of 2^n unique output lines. It does not have memory, so it is not a sequential circuit.
- **Adder** is a combinational circuit used for adding binary numbers. It does not store any state information, hence it is not a sequential circuit.

Positive Marks: 4.00

Negative Marks: 1.00

[51]: What is the time complexity of finding the maximum number in an unsorted array of n elements?

Option 1:

$O(1)$

Option 2:

$O(\log n)$

Option 3:

$O(n)$

Option 4:

$O(n^2)$

SolutionHeading: Full Solution

Text:

The correct answer is **$O(n)$** .

Key Points

- The time complexity of finding the maximum number in an unsorted array of n elements is **$O(n)$** .
- This is because you need to check each element of the array to ensure you have found the maximum value.
- In the worst-case scenario, every element must be compared, leading to a linear time complexity of $O(n)$.

Additional Information

- **$O(1)$** : This represents constant time complexity, which means the operation takes the same

amount of time regardless of the size of the input. This is not applicable here because we need to scan all elements in the array.

- **$O(\log n)$** : This represents logarithmic time complexity, which usually applies to operations in sorted arrays or binary trees. It is not applicable for finding the maximum in an unsorted array.
- **$O(n^2)$** : This represents quadratic time complexity, which is typically seen in algorithms like bubble sort or selection sort. It is much higher than necessary for finding the maximum in an unsorted array.

Positive Marks: 4.00

Negative Marks: 1.00

[52]: Which of the following sort algorithms has execution time that is least dependent on initial ordering of the input?

Option 1:

Insertion sort

Option 2:

Quick sort

Option 3:

Merge sort

Option 4:

Selection sort

SolutionHeading: Full Solution

Text:

Option_1: Insertion Sort

In Insertion sort, if the array is already sorted, then it takes $O(n)$ time and if it is sorted in decreasing order, then it takes $O(n^2)$ time to sort the array.

Option_2: Quick Sort

In Quick sort, if the array is already sorted whether in decreasing or in non-decreasing order, then it takes $O(n^2)$ time.

Option_3 – Merge Sort

Merge sort gives time complexity of $O(n \log n)$ in every case, be it best, average or worst. In merge sort, performance is affected least by the order of input sequence.

Option 4: Selection Sort

For selection sort, the best- case and worst-case performance of Selection is $O(n^2)$ only. But if the array is already sorted then less swaps are required.

Positive Marks: 4.00

Negative Marks: 1.00

[53]: Which of the following ways of organizing the directory allows cycle between two nodes?

Option 1:

Two level directory

Option 2:

Tree structure directory

Option 3:

Acylic graph directory

Option 4:

General graph directory

SolutionHeading: Full Solution

Text:

Directory: It can be defined as a listing of the related files on the disk. It stores attributes of the files as well.

Two-level directories: In this directory structure, users have their own User Files Directories (UFD) and these UFDs contain only the files of a single user. It has one master directory, user directories and then files in those user directories. Each user can have their own separate directory. It does not allow cycles.

Tree Structure Directories: It is an extended version of a two-level directory structure. It allows to create sub-directories and organize files accordingly. It is a very generalized structure since full path names can

be given. Grouping capability can also be found. It does not allow cycles.

Acyclic Graph Directories: In this directory structure, unlike the tree structure, allows directories to share sub-directories and files, that is, two or more directory entries can point to the same subdirectory. Hence, changes made to one user are visible to all others sharing the same file. It does not allow cycles.

General Graph Directories: In this directory structure, cycles are allowed within a directory structure where multiple directories can be derived from more than one parent directory. It allows links to be created back up the chain of directory structures, hence, it allows cycles.

Positive Marks: 4.00

Negative Marks: 1.00

[54]: Which of the following option has linear data structure?

Option 1:

Turnery Tree

Option 2:

binary Tree

Option 3:

Stack

Option 4:

Graph

SolutionHeading: Full Solution

Text:

The correct answer is Stacks

Key Points

A linear data structure is a structure wherein data elements are adjacent to each other, and each data element connects to its previous and next element. The elements are arranged in a sequential order where they have a predecessor and a successor, except the first and last elements. Examples of linear

data structures include Arrays, Linked Lists, Stacks, and Queues.

- Arrays: In an array, data is stored in a contiguous block of memory. Each element in the array can be accessed directly using its index.
- Linked Lists: Linked lists also contain a collection of data elements where each element points to the next element in the sequence. Data elements aren't stored in continuous memory like arrays.
- Stacks: A stack is a linear data structure that follows a particular order in which operations are performed. The order is LIFO (Last In, First Out) meaning that the last element added to the stack will be the first one to be removed.
- Queues: It's another linear data structure that follows the FIFO (First In, First Out) principle. The first element added to the queue will be the first one to be removed.

So, among the stack is a linear data structure.

Additional Information

A non-linear data structures are structured in a way where each data element can connect to two or more other data elements and a hierarchical relationship exists among the data elements. Examples include Trees, Graphs, B-trees, etc.

- Tree: A tree is a non-linear data structure that emulates a tree (in biology), with a set of linked nodes. It starts at a top node (the root) and proceeds downwards to the leaf nodes with a connection between each parent node and its children nodes.
- Graph: A graph is also a non-linear data structure. In a graph, data elements are interconnected with

each other, where each node can connect with any other nodes in the graph.

- Binary Tree: This is a type of tree where each node can have at most 2 children: a left child and a right child.

Positive Marks: 4.00

Negative Marks: 1.00

[55]: 15th complement of Hexadecimal number (A10)₁₆ is:

Option 1:

(5EF)₁₆

Option 2:

(6F0)₁₆

Option 3:

(6FF)₁₆

Option 4:

(5E0)₁₆

SolutionHeading: Full Solution

Text:

The correct answer is (5FE)₁₆

Key Points

- To find the 15's complement of a hexadecimal number, you need to take each digit and subtract it from 15 ('F' in hexadecimal).

EXPLANATION:

- The hexadecimal number given is (A10)₁₆.
- The 15's complement of A is F - A = 5.
- The 15's complement of 0 is F - 1 = E.
- The 15's complement of 1 is F - 0 = F.
- So, the 15's complement of (A01)₁₆ is (5EF)₁₆.

Therefore, the correct answer is option 1) (5EF)₁₆.

Positive Marks: 4.00

Negative Marks: 1.00

[56]: Consider an array A[2 : 8, -4 : 1, 6 : 10]. Base

address of array A is = 400 and size of each element = 2 word per memory cell. What is the address of A[5, -1, 8] in row major form?

Option 1:

628

Option 2:

630

Option 3:

626

Option 4:

614

SolutionHeading: Full Solution

Text:

Key Points

Three-dimensional array A[b1...u1, b2...u2, b3...u3]

Here b is base value, u max value

number of planes = $u_1 - b_1 + 1 = r_1$

number of rows = $u_2 - b_2 + 1 = r_2$

number of columns = $u_3 - b_3 + 1 = r_3$

Row major order of array A[i, j, k] = $L_0 + \{(i - b_1) \times r_2 \times r_3 + (j - b_2) \times r_3 + (k - b_3)\} \times c$

L_0 = Base address

c = size of each element

$r_1 = 8 - 2 + 1 = 7$

$r_2 = 1 - (-4) + 1 = 6$

$r_3 = 10 - 6 + 1 = 5$

$A[5, -1, 8] = 400 + [(5 - 2) \times 6 \times 5 + (-1 - (-4)) \times 5 + (8 - 6)] \times 2$

$= 400 + [90 + 15 + 2] \times 2$

$= 614$

Hence the correct answer is 614.

Positive Marks: 4.00

Negative Marks: 1.00

[57]: Consider a system that has 4K pages of 512 bytes

in size in the logical address space. The number of bits of logical address?

Option 1:

21

Option 2:

20

Option 3:

19

Option 4:

18

SolutionHeading: Full Solution

Text:

The correct answer is **option 1.**

Concept:

The given data,

Number of pages = 4 K

Page size PS= 512 B = 2^9 Bytes

Number of pages = Logical address space / Page size

Logical address space = Number of pages X Page size

Logical address space = 4 K X 2^9 Bytes

Logical address space = 2^{12} X 2^9 Bytes

Logical address space = 2^{21} Bytes

Logical address bits or Virtual address bits = 21

Hence the correct answer is 21.

Positive Marks: 4.00

Negative Marks: 1.00

[58]: _____ computer program loads a library into memory, retrieves the variable and function addresses, executes the functions, and unloads the program from memory when it runs.

Option 1:

Static loading Dynamic

Option 2:

Option 3:

Dynamic loading

Option 4:

Overlays

The correct answer is **option 3.**

Concept:

Dynamic loading:

Dynamic loading is the process by which a computer programme loads a library into memory, obtains the variable and function addresses, performs the functions, and then unloads the programme from memory. Use of this technique is common while implementing plugins.

In dynamic loading, a system library or routine is loaded at run-time. When a dependent library or routine is linked dynamically at run-time, it is called dynamic linking.

Hence the correct answer is Dynamic loading.

Additional Information Dynamic linking:

At dynamic linking, code is compiled and linked into a format that programmes may load both at link time and during run time. They differ from standard object files because of this.

Overlays:

Overlaying is a programming method that allows programs to be larger than the computer's main memory.

Positive Marks: 4.00

Negative Marks: 1.00

[59]: Which of the following is not a contiguous memory management technique?

Option 1:

Option 2:

Option 3:

Option 4:

SolutionHeading: Full Solution

Text:

The correct answer is **option 3**.

Concept:

Contiguous memory management:

A traditional memory allocation model is contiguous memory allocation. In this case, a system gives a process access to successive memory blocks, or memory blocks with consecutive addresses. One of the earliest types of memory allocation is contiguous memory allocation.

There are three types of contiguous memory management:

- . Over lays
- . Partition
- . Buddy system

Explanation:

Paging is a non-contiguous memory allocation technique that divides secondary and main memory into equal-size divisions. Secondary memory partitions are referred to as pages, whilst major memory partitions are referred to as frames.

Hence the correct answer is Paging.

Positive Marks: 4.00

Negative Marks: 1.00

[60]: Which of the following helps the OS in allocating a physical memory location to a logical pointer?

Option 1:

Option 2:

Option 3:

Option 4:

SolutionHeading: Full Solution

Text:

Loading: This process is responsible to place a load module in the main memory at some starting address. A library of functions will be loaded into the memory which will be required during the execution.

Linking: In this process, a collection of object modules will be taken as input and a load module consisting of an integrated set of programs and data modules is passed to the loader. In each object module, there may be a symbolic reference to a location in other modules.

Address Binding: Address binding helps in the allocation of the physical memory location to the logical pointer by associating a physical address with the logical address or virtual address. In this part of the computer, memory management is performed by the operating system on the behalf of applications that need access to the memory.

Positive Marks: 4.00

Negative Marks: 1.00

[61]: The aim of creating page replacement algorithms is to :

Option 1:

replace pages faster

Option 2:

increase the page fault rate

Option 3:

decrease the page fault rate

Option 4:

to allocate multiple pages to processes

SolutionHeading: Full Solution

Text:

In Operating System, a page replacement algorithm is needed to decide which page needs to be replaced when a new page comes in.

What is Page fault-

A page fault happens when a running program accesses a memory page not loaded in physical memory. Since actual physical memory is much smaller than virtual memory, page faults happen.

In case of a page fault, Operating System might have to replace one of the existing pages with the newly needed page.

Page faults are always bad for the operating system, and without a page replacement algorithm, it will take much time to search that particular page in the main memory.

Different Page replacement algorithm-

- **FIFO(First in First out)**-In this algorithm, the operating system keeps track of all pages in the memory in a queue, the oldest page is in the front of the queue. When a page needs to be replaced page in the front of the queue is selected for removal.
- **OPTIMAL ALGORITHM**-In this algorithm, pages are replaced which would not be used for the longest duration of time in the future.
- **LRU(Least Recently Used)**-In this algorithm, the page will be replaced which is least recently used.

Let's analyze the options one by one-

Option 1- we need a page replacement algorithm, here the aim is not to replace pages faster.

Option 2- It's required to decrease the page fault rate, not to increase page fault rate because the page fault rate is bad for the operating system.

Option 3- This is the answer as explained above.

Option 4- All process does not require multiple pages.

Positive Marks: 4.00

Negative Marks: 1.00

[62]: In Round Robin CPU scheduling, as the time quantum is increased, the average turn around time

Option 1:

increases

Option 2:

decreases

Option 3:

remains constant

Option 4:

varies irregularly

SolutionHeading: Full Solution

Text:

The correct answer is **varies irregularly**.

Key Points

- The Round-Robin is known as a CPU scheduling algorithm.
- The Round-Robin method mainly used the time-sharing method. The time frame for which a method or task is permitted to run in a pre-emptive technique is known as time quantum.
- In Round Robin CPU scheduling, as the time quantum is increased, the average turn-around

time varies irregularly.

Thus the correct answer is *varies irregularly*.

Additional Information

- The main characteristics of the Round-Robin method are-
 - It is very easy to implement in the system.
 - In this process, time is shared equally for each process.
- The main advantage of the Round Robin Scheduling is time slicing. If any new process is created, then it is always added at the end of the queue.
- The main disadvantage of the Round Robin Scheduling is waiting time is more than the response time. It has very low throughput.

Positive Marks: 4.00

Negative Marks: 1.00

[63]: Which of the following searching technique does not get affected on increasing the size of search list.

Option 1:

Binary Search

Option 2:

Linear Search

Option 3:

Search by Hashing

Option 4:

None of the above

SolutionHeading: Full Solution

Text:

The correct option is **Search by Hashing**.

CONCEPT:

In linear search, every element of a given list is compared one by one with the given key without skipping any element.

It is useful when we need to search for an item in a small unsorted list, but the **time taken to search the list increases as the size of the list increases**.

For example, consider a list of length 5 and the key element is present at the end of this list. Number of comparisons required to search the key element = size of list i.e 5

If we increase the size of the same list (say 15) and the key element is present at the end of this list. Number of comparisons required to search the key element = size of list i.e 15

In binary search, the key to be searched is compared with the middle element of a sorted list, If the element at the middle position:

- i) Matches the key the search is successful
- ii) Is greater than the key then the key element may be present in the left part of the list
- iii) Is smaller than the key, then the key element may be present in the right part of the list

This process continues till the element is found or the list is traversed completely.

Thus the time taken to search the list increases as the size of the list increases. But it will not be as large as the time required by linear search

Hash-based searching requires only one key comparison to find the position of a key, provided every element is present at its designated position decided by a hash function.

For example, consider a list of length 5 and the hash function: $h(\text{element}) = \text{element} \% \text{size}(\text{list})$

A hashing function is a mathematical function that generates unique results for each unique value supplied to the hash function in constant time.

For example: Consider a list of length 5 and if we want to search for key = 12, the index returned by the hash function is $h(12) = 12 \% 5 = 2$ and requires only one key comparison to find the key at that index

Similarly increasing the size of the list (say 15) and if we want to search for key = 12, the index returned by the hash function is $h(12) = 12 \% 15 = 12$ and requires only one key comparison to find the key at that index

Thus it is independent of the length of the list.

Positive Marks: 4.00

Negative Marks: 1.00

[64]: Convert the hexadecimal number $(A72E.BF85)_{16}$ into the octal number system.

Option 1:

$(123456.552024)_8$

Option 2:

$(234156.577024)_8$

Option 3:

$(123456.577024)_8$

Option 4:

$(112346.572024)_8$

SolutionHeading: Full Solution

Text:

The correct answer is $(123456.577024)_8$

Concept:

Convert the integer part (left of the decimal point) to binary:

A = 1010

7 = 0111

2 = 0010

E = 1110

Concatenating these binary values: 1010 0111 0010 1110

Convert the fractional part (right of the decimal point) to binary:

B = 1011

F = 1111

8 = 1000

5 = 0101

Concatenating these binary values: 1011 1111 1000 0101

Group the binary digits into sets of three, starting from the decimal point:

Integer part: 1 010 011 100 101 110

Fractional part: 101 111 111 000 010 100

Convert each group of three binary digits to octal:

Integer part: 1 2 3 4 5 6

Fractional part: 5 7 7 0 2 4

(123456.577024)₈

Positive Marks: 4.00

Negative Marks: 1.00

[65]: A combinational logic circuit which is used when it is desired to send data from two or more source through a single transmission line is known as

Option 1:

encoder

Option 2:

decoder

Option 3:

multiplexer

Option 4:

demultiplexer

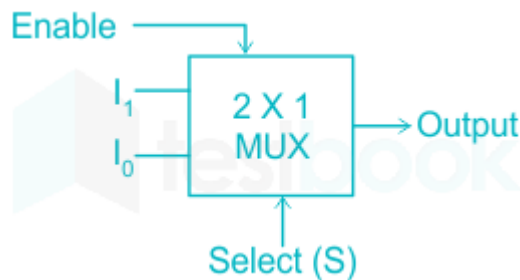
SolutionHeading: Full Solution

Text:

Multiplexer:

- Combining one or more signals and transmitting on a single channel
- It consists more than one input line and only one output line and more than one select line

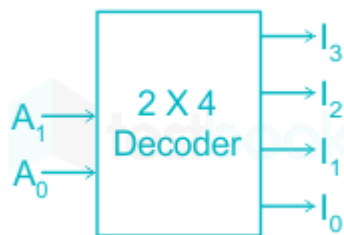
2 X 1 MUX



Decoder:

- It consists of n input lines and maximum of 2^n output lines.
- Outputs of the decoder are nothing but the min terms of n input variables.

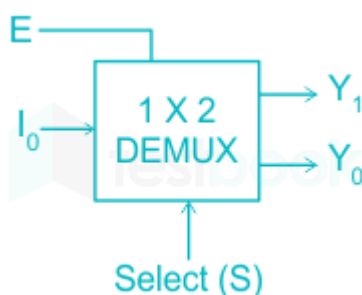
2 X 4 Decoder



Encoder:

- Reverse operation of decoder.
- It consists of 2^n input lines and n output lines.
- Enable signal is optional to represent.

$2^2 \times 2$ Encoder



Hence option 3 is the correct answer.

[66]: Assume that there are 3 frames and consider the following reference string:

D, C, B, A, A, E, B, A, E, D, C, B

Count the number of page faults that occur when FIFO Page Replacement Algorithm is used.

Option 1:

6

Option 2:

7

Option 3:

8

Option 4:

9

SolutionHeading: Full Solution

Text:

Page Fault: It is a situation in which a running process tries to access a memory page that is mapped in the virtual address but is not loaded in the physical memory. Each page table entry has a valid bit which indicates whether the corresponding page is in the memory or not.

FIFO Page Replacement: As soon as the page fault occurs, a page in the physical memory is to be selected to be replaced with the page required at the moment. So in this algorithm, that page is selected which has been in the memory for the longest time, that is, the page that entered in the memory first will be replaced with the required page first.

Calculation: D,C,B,A,A,E,B,A,E,D,C,B

D	C	B	A	A	E	B	A	E	D	C	B
		B	B	B	B	B	B	B	D	D	D
	C	C	C	C	E	E	E	E	E	E	B
D	D	D	A	A	A	A	A	A	A	C	C
F	F	F	F		F				F	F	F

. F = Page Faults

- In the above table, first row shows which column arrived. It is same as the string given in the question.
- Each cell of the table (2nd, 3rd and 4th row) is a memory frame.
- In the last row, F represents that the page fault has occurred. Black cell shows Page is already in the memory and hence, no fault at that moment.

Positive Marks: 4.00

Negative Marks: 1.00

[67]: Which of the following is an advantage of fragmentation occurring in the memory?

Option 1:

Fast Data Writes

Option 2:

Fewer Failures

Option 3:

Stable Optimisation

Option 4:

All of the above

SolutionHeading: Full Solution

Text:

Fragmentation: It is a situation or problem occurring in operating system where many processes are loaded and unloaded in the memory which results in more and more fragmented pieces of memory. And then a time comes when the total available memory is more than the required memory for a process but since it is not contiguous, it is useless.

Advantages of Fragmentation:

1. **Fast Data Writes:** With fragmentation present in the memory, data writing turns out to be faster as in the other cases, reorganising the data storage to enable contiguous data writes wastes a lot of time.

2. **Fewer Failures:** There has to be sufficient space in the memory so that when data writes are done, write shouldn't fail. This probability is much higher in systems which don't support fragmentation.
3. **Stable Optimisation:** Sometimes, processes arrive in such a way that fragmented systems potentially make better use of a storage device by utilising every available storage block.

Disadvantages of Fragmentation:

1. **Need of regular defragmentation:** With time, the performance of the storage device will reduce significantly which will further require for time consuming defragmentation processes.
2. **Slower Read Times:** If the storage device or memory becomes more and more fragmented, it will take equivalently more time to read the non-sequential data from the memory.

Positive Marks: 4.00

Negative Marks: 1.00

[68]: The cost of 5 notebooks and 4 pens is ₹ 142. The cost of 7 notebooks and 3 pens is ₹ 28 less than the cost of 9 notebooks and 1 pen. What is the cost of 4 notebooks and 5 pens?

Option 1:

150

Option 2:

136

Option 3:

144

Option 4:

128

SolutionHeading: Full Solution

Text:

Given:

5 notebooks + 4 pens = ₹142

7 notebooks + 3 pens = 9 notebooks + 1 pen - ₹28

Formula used:

Form and solve linear equations.

Calculation:

Let the cost of one notebook be 'n' and the cost of one pen be 'p'.

Then,

$$5n + 4p = 142 \dots (1)$$

Also,

$$7n + 3p = 9n + p - 28$$

$$7n + 3p - 9n - p = -28$$

$$-2n + 2p = -28$$

$$-n + p = -14$$

$$p = n - 14 \dots (2)$$

Substitute equation (2) into equation (1):

$$5n + 4(n - 14) = 142$$

$$5n + 4n - 56 = 142$$

$$9n = 142 + 56$$

$$9n = 198 = 198 / 9$$

$$n = 22$$

Substitute $n = 22$ into equation (2):

$$p = 22 - 14$$

$$p = 8$$

Now the cost of 4 notebooks and 5 pens:

$$4n + 5p = 4 \times 22 + 5 \times 8$$

$$4n + 5p = 88 + 40$$

$$4n + 5p = 128$$

$\therefore$ The cost of 4 notebooks and 5 pens is ₹128.

[69]: The letters of the word "COCHIN" are permuted and all the permutations are arranged in alphabetical order as in an English dictionary. The number of words that appear before the word "COCHIN" is

Option 1:

48

Option 2:

96

Option 3:

192

Option 4:

360

SolutionHeading: Full Solution

Text:

Calculation

The number of words starting with CC = $4!$

The number of words starting with CH = $4!$

The number of words starting with CI = $4!$

The number of words starting with CN = $4!$

COCHIN is the first word in the list of words beginning with CO.

: Number of words that appear before the words
COCHIN = 96

Hence option 2 is correct

Positive Marks: 4.00

Negative Marks: 1.00

[70]: The sum of the first 11 terms of the A.P. 3, 8, 13, is

Option 1:

380

Option 2:

318

Option 3:

308

Option 4:

388

SolutionHeading: Full Solution

Text:

Given:

First term (a) = 3

Common difference (d) = 5

Number of terms (n) = 11

Formula Used:

Sum of the first n terms of an A.P. (S_n) = $n/2 \times (2a + (n-1)d)$

Calculation:

$$S_{11} = 11/2 \times (2 \times 3 + (11-1) \times 5)$$

$$\Rightarrow S_{11} = 11/2 \times (6 + 50)$$

$$\Rightarrow S_{11} = 11/2 \times 56$$

$$\Rightarrow S_{11} = 11 \times 28$$

$$\Rightarrow S_{11} = 308$$

The sum of the first 11 terms is 308.

Answer: 3) 30

Positive Marks: 4.00

Negative Marks: 1.00

[71]: Select the option that is related to the fifth letter-cluster in the same way as the second letter-cluster is related to the first letter-cluster and the fourth letter-cluster is related to the third letter-cluster.

MOBILE : CPNFMJ :: PHONES : PIQTFO ::
CHARGE : ?

Option 1:

BIDFSH

Option 2:

BIDFHS

Option 3:

BIFDHS

Option 4:

BIEFHS

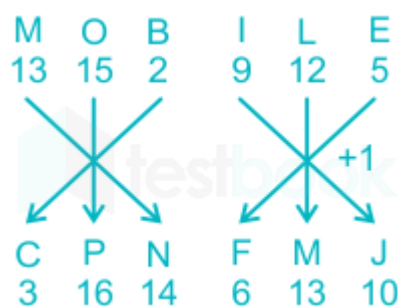
SolutionHeading: Full Solution**Text:**

The position of the letters according to the English alphabet series:

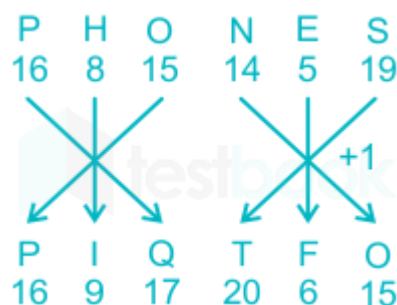
Alphabets	A	B	C	D	E	F	G	H	I	J	K	L	M
Positional value	1	2	3	4	5	6	7	8	9	10	11	12	13
Positional value	26	25	24	23	22	21	20	19	18	17	16	15	14
Alphabets	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

The logic followed here is:

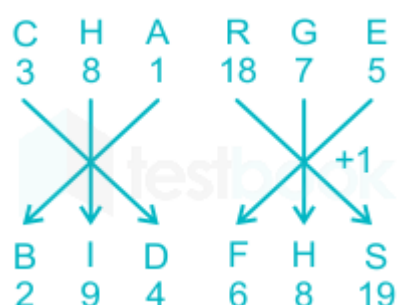
. MOBILE : CPNFMJ



. PHONES : PIQTFO



. Similarly, **CHARGE : BIDFHS**



Hence, the correct answer is '**BIDFHS**'.

[72]: **Directions:** Below a question is given with two statements (I) and (II). These statements may be either independent causes or may be effects of independent causes or a common cause. One of these statements may be the effect of the other statement. Read both the statements and decide which of the following answer choice correctly depicts the relationship between these two statements.

(A) Ranbir has given many competitive exams.

(B) Ranbir knows the pattern of various competitive exams.

Option 1:

Statement (A) is the cause and statement (B) is its effect

Option 2:

Statement (B) is the cause and statement (A) is its effect

Option 3:

Both statements (A) and (B) are independent causes

Option 4:

Both statements (A) and (B) are effects of independent causes

SolutionHeading: Full Solution

Text:

Ranbir knows the pattern of various competitive exams as he has given many competitive exams.

Hence, Statement (A) is the cause and statement (B) is its effect.

[73]: Which fraction comes next in the sequence

$$\frac{2}{3}, \frac{5}{8}, \frac{8}{13}, \frac{11}{18}, ?$$

Option 1:

$$\frac{14}{23}$$

Option 2:

$$\frac{15}{23}$$

Option 3:

$$\frac{16}{23}$$

Option 4:

$$\frac{17}{23}$$

SolutionHeading: Full Solution

Text:

The Correct answer is Option 1.

Key Points

Given Sequence:

$$\frac{2}{3}, \frac{5}{8}, \frac{8}{13}, \frac{11}{18}$$

Identify the Numerator Pattern

Numerators: 2, 5, 8, 11, ?

Each numerator increases by 3.

$$2 + 3 = 5$$

$$5 + 3 = 8$$

$$8 + 3 = 11$$

Next numerator: $11 + 3 = 14$

Identify the Denominator Pattern

Denominators: 3, 8, 13, 18, ?

Pattern: Each denominator increases by 5.

$$3 + 5 = 8$$

$$8 + 5 = 13$$

$$13 + 5 = 18$$

Next denominator: $18 + 5 = 23$

Find the Next Fraction

The next fraction is: $14 / 23$

Hence Option 1 is correct.

Positive Marks: 4.00

Negative Marks: 1.00

[74]: Select the option that is related to the third word in the same way as the second word is related to the first word.

(The words must be considered as meaningful English words and must not be related to each other based on the number of letters/number of consonants/ vowels in the word.)

Summon : Call :: Disperse : ?

Option 1:

Surround

Option 2:

Remove

Option 3:

Spread

Option 4:

Disappear

SolutionHeading: Full Solution

Text:

The logic followed here is:

Logic: **Synonyms**

Here,

Summon : Call → **Call** is a synonym of **Summon**.

Similarly,

Disperse : Spread → **Spread** is a synonym of **Disperse**.

Hence, the correct answer is "Option 3".

[75]: A son-in-law remembers that his mother-in-law's birthday definitely falls before the twentieth of August, while his wife remembers that she celebrated her mother's birthday definitely after the eighteenth of August and before the twenty-second of August. Considering both of them correct, tell on which date is the birthday of that person's mother-in-law?

Option 1:

19 August

Option 2:

20 August

Option 3:

18 August

Option 4:

21 August

SolutionHeading: Full Solution

Text:

Given:

The son-in-law remembers the birthday is **before August 20**.

The wife remembers the birthday is **after August 18 and before August 22**

Analyze the son's statement.

The son-in-law is certain that the birthday occurs before August 20.

Possible dates based on his memory are: **August 18, 19**.

Analyze the wife's statement.

The wife is certain that the birthday occurs **after August 18 and before August 22**.

Possible dates based on her memory are: **August 19, 20, 21**.

The common date satisfying both statements is **August 19**.

Thus, The mother-in-law's birthday is on **August 19**.
Hence, the correct answer is "**Option 1**".

Positive Marks: 4.00

Negative Marks: 1.00